

Figure 1: Architecture of Docker Swarm

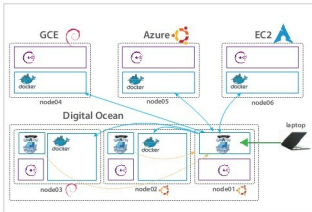


Figure 2: Multi-cloud application delivery using Docker Swarm

```
docker --
Installing new version of config file /etc/init.d/docker --
Installing new version of config file /etc/init/docker.conf --
Processing triggers for ureadahead (0.100.0-10) --
Processing triggers for systemd (225-1ubuntu9) --
+ sh -c docker version
Client:
Version: 1.9.1
API version: 1.21
Go version: go1.4.2
Git commit: a34a1d5
Built: Fri Nov 20 13:20:08 UTC 2015
OS/Arch: linux/amd64
Server:
Version: 1.9.1
API version: 1.21
Go version: go1.4.2
Git commit: a34a1d5
Built: Fri Nov 20 13:20:08 UTC 2015
OS/Arch: linux/amd64
```

If you would like to use Docker as a non-root user, you should now consider adding your user to the 'docker' group with something like what follows:

```
sudo usermod -aG docker your-user
```

No	Machine Name	IP Address
1	Swarm Master Node	10.240.0.2
2	Swarm Agent Node 1	10.240.0.3
3	Swarm Manager Node	10.240.0.5
4	Swarm Agent Node 2	10.240.0.5

Table 1



Note: You will have to log out and log in again for this to take effect.

Starting the Docker daemon

The Docker daemon is a self-sufficient runtime for Docker containers. Let's configure and start Docker Engine so that it listens on port 2375, as shown below:

```
root@dockerhost-1 ~]# docker daemon -H tcp://0.0.0.0:2375 -d &
[1] 11516
```

Setting up Docker Swarm

Run the following command to install Docker Swarm on the master node and create a Swarm token, which is the heart of the Docker Swarm cluster configuration, as follows:

```
root@docker-1:~# docker run -rm swarm create
7733f838d176809cb2f2d24eb34ce78c
```

As shown in the output above, it created a token ID starting with 7733f838.

Setting up the Swarm agent node 1: [10.240.0.3]

Ensure that Docker 1.9.x is installed on Agent Node 1. Also, the Docker daemon needs to be running, following steps similar to those followed during the master node set-up.

```
#docker run -d swarm join --addr=10.240.0.3:2375 token:/
/7733f838d176809cb2f2d24eb34ce78c 560e58a76ef235c8d74f
ffb2c680149111b8bbe687f 1e0e164cd7c095da5d9f33
```

Setting up Swarm manager node: [10.240.0.5]

Swarm manager helps you to start managing your Swarm cluster. Let's set up a Swarm manager on a separate node (though it can be set up on a master node too).

```
root@docker-3:~# docker run -d -p 7000:2375 swarm manage toke
n://7733f838d176809cb2f2d24eb34ce78cb26a2bbb336e26e0cd6ac5d61
b90d7a9d9f74a2d072f573b6890b9b0f6e470f
```

Please remember that 7000 is the Swarm manager port which is being set.

Now you can see the Swarm cluster details:

```
root@docker-3:~# docker -H tcp://10.240.0.5:7000 info
Containers: 0
```